



Systematic Layout Planning (SLP) tutorial

March 2026
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TUTORIAL : 6 departments different sizes

6.8 with 6 departments starting at From-To chart

6.8a From-To Chart

Dept.	A	B	C	D	E	F
A	—	13,320	1,800	0	0	0
B	0	—	11,400	6,600	4,920	5,400
C	0	6,600	—	2,400	4,200	3,600
D	0	3,000	1,200	—	5,040	960
E	0	5,400	7,800	1,200	—	5,160
F	0	0	0	0	0	—

Flow-Between Chart

6.8

Dept.	A	B	C	D	E	F
A	–	13,320	1,800	0	0	0
B		–	18,000	9,600	10,320	5,400
C			–	3,600	12,000	3,600
D				–	6,240	960
E					–	5,160
F						–

Determine rankings....let us see what you do...analytical calculation vs gut feel vs logical groupings

Flow-Between Chart

6.8

Dept.	A	B	C	D	E	F
A	—	13,320	1,800	0	0	0
B		—	18,000	9,600	10,320	5,400
C			—	3,600	12,000	3,600
D				—	6,240	960
E					—	5,160
F						—

18000/5 = interval sizes of 3600 resulting in:

14400-18000	A = 1
10800-14400	E = 2
7200-10800	I = 2
3600-7200	O = 5
0-3599	U = 2

Arbitrary intervals (what they did) resulted in:

13000-18000	A = 2
9000-12000	E = 3
5000-9000	I = 3
3000-4000	O = 2
0-2000	U = 2

Activity Relationship Chart

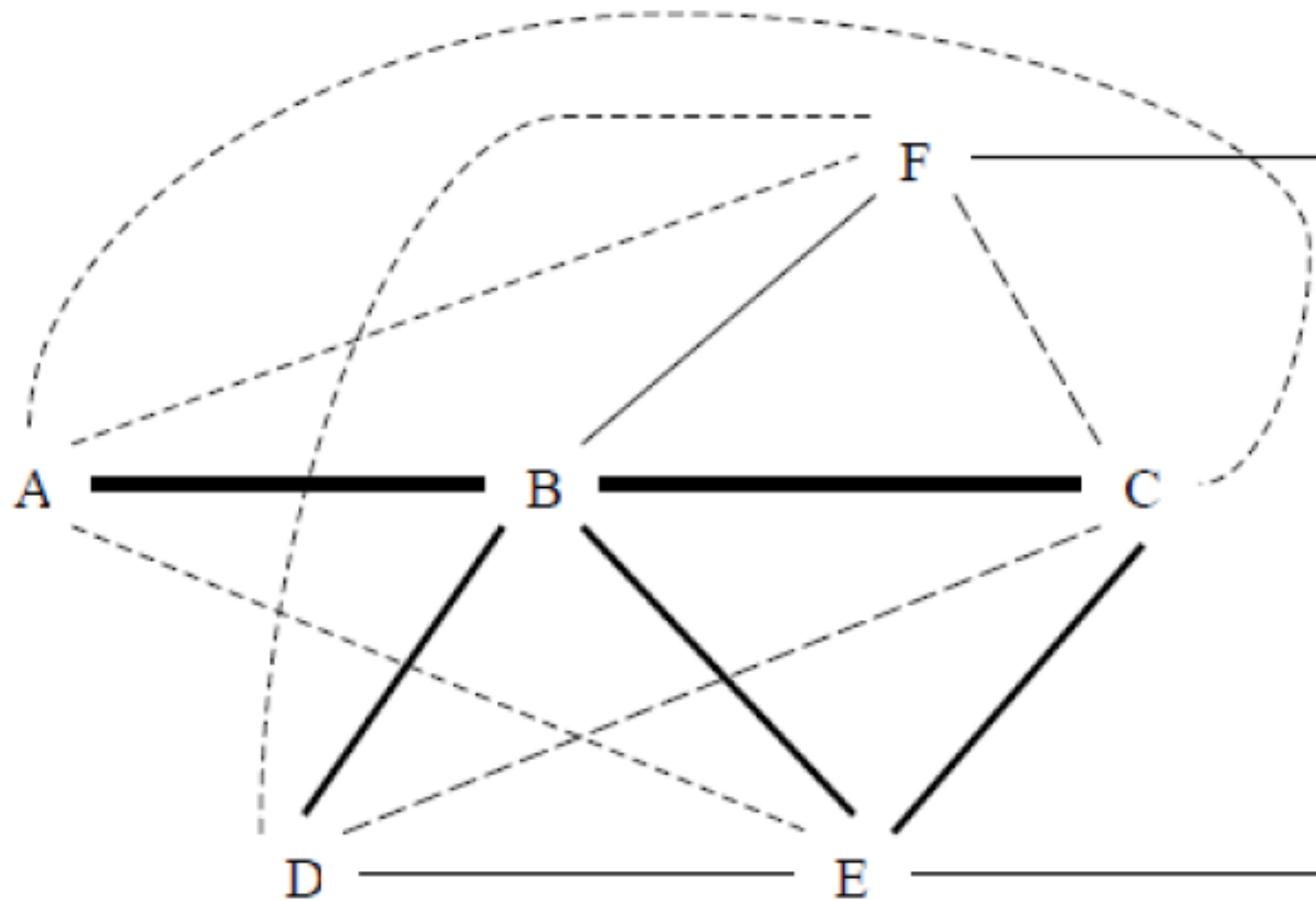
Dept.	A	B	C	D	E	F
A	—	A	U	U	U	U
B		—	A	E	E	I
C			—	O	E	O
D				—	I	U
E					—	I
F						—

18000/5 = interval sizes of 3600 resulting in:

- A = 1
- E = 2
- I = 2
- O = 5
- U = 2

Arbitrary intervals (what they did) resulted in:

- A = 2
- E = 3
- I = 3
- O = 2
- U = 2



Block sizes:

A = 40x40

B = 45x45

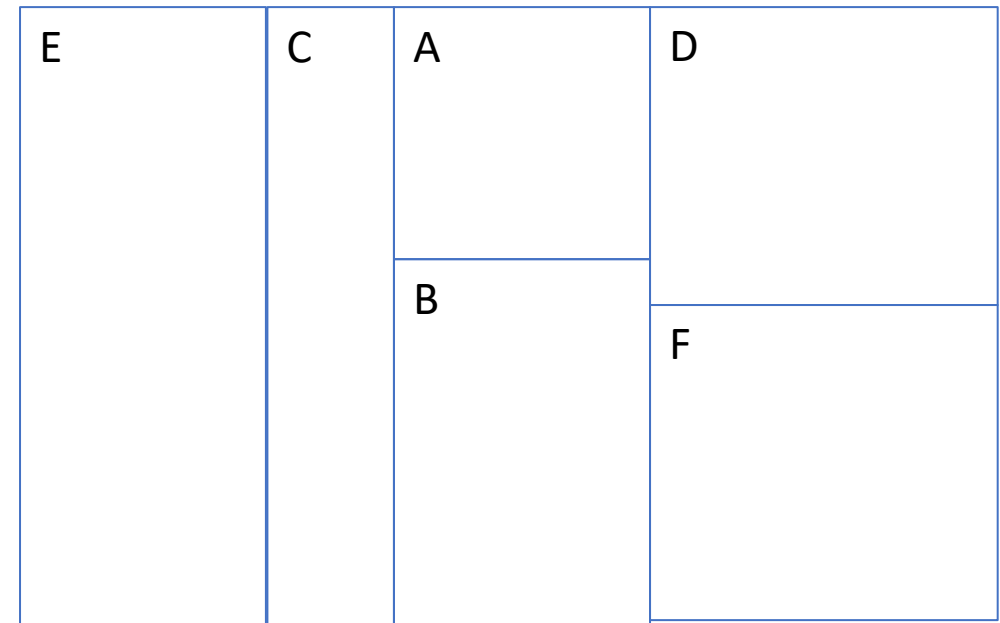
C = 30x30 (smallest)

D = 50x50

E = 60x60 (largest)

F = 50x50

BLOCK LAYOUT - a possible solution



TUTORIAL : 8 departments same size

Q11 from Chase

There are many ways to approach this problem. This is just what made sense to me.
Firstly I thought it would be nice to have a relationship diagram to visualise what is happening.
To do this I needed to tabulate volumes and then aggregate these between departments.

DEPARTMENTS								
	1	2	3	4	5	6	7	8
1	—	20						
2	15	—	25				4	
3		5	—	40	5			
4			5	—	10			
5	1			20	—	30		
6						—	20	
7				3			—	10
8							5	—

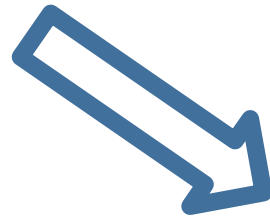
Flow	Volume
1-2	20
2-1	15
2-3	25
2-7	4
3-2	5
3-4	40
3-5	5
4-3	5
4-5	10
5-1	1
5-4	20
5-6	30
6-7	20
7-4	3
7-8	10
8-7	5



Aggregated flows	Aggregated volumes
1-2-1	35
2-3-2	30
3-4-3	45
3-5-3	5
4-5-4	30
2-7-2	4
5-1-5	1
5-6-5	30
6-7-6	20
7-4-7	3
7-8-7	15

I could do this as the question said that the direction of flow is not important.

Aggregated flows	Aggregated volumes
1-2-1	35
2-3-2	30
3-4-3	45
3-5-3	5
4-5-4	30
2-7-2	4
5-1-5	1
5-6-5	30
6-7-6	20
7-4-7	3
7-8-7	15



Aggregated flows	Aggregated volumes	Ranking
1-2-1	35	4
2-3-2	30	4
3-4-3	45	5
3-5-3	5	1
4-5-4	30	4
2-7-2	4	1
5-1-5	1	1
5-6-5	30	4
6-7-6	20	3
7-4-7	3	1
7-8-7	15	2
Ranking:		
	1: 0-9	
	2: 10-19	
	3: 20-29	
	4: 30-39	
	5: 40-49	

DEPARTMENTS								
	1	2	3	4	5	6	7	8
1	—	20						
2	15	—	25				4	
3		5	—	40	5			
4			5	—	10			
5	1			20	—	30		
6						—	20	
7				3			—	10
8							5	—

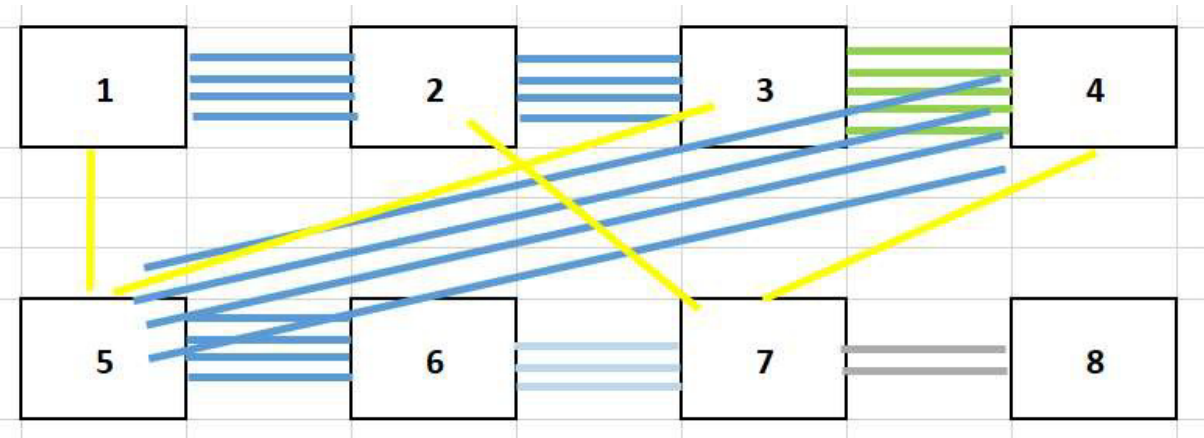
I then decided to rank the volume of flow so that I could draw a relationship diagram.

I just made up the ranking table based on the volumes that I could see in the table.

Aggregated flows	Aggregated volumes	Ranking
1-2-1	35	4
2-3-2	30	4
3-4-3	45	5
3-5-3	5	1
4-5-4	30	4
2-7-2	4	1
5-1-5	1	1
5-6-5	30	4
6-7-6	20	3
7-4-7	3	1
7-8-7	15	2
Ranking:	1: 0-9	
	2: 10-19	
	3: 20-29	
	4: 30-39	
	5: 40-49	

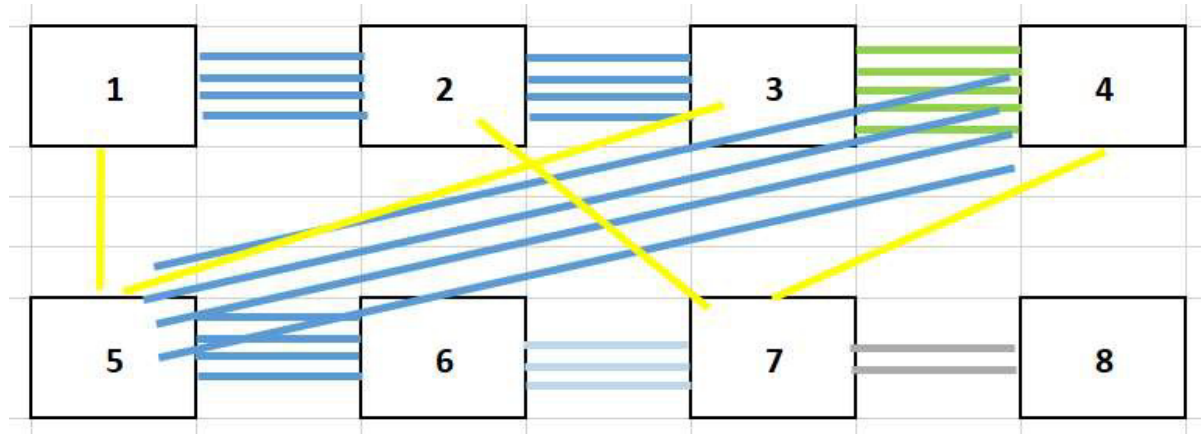
1	2	3	4
5	6	7	8

DEPARTMENTS								
	1	2	3	4	5	6	7	8
1	—	20						
2	15	—	25				4	
3		5	—	40	5			
4			5	—	10			
5	1			20	—	30		
6						—	20	
7				3			—	10
8							5	—

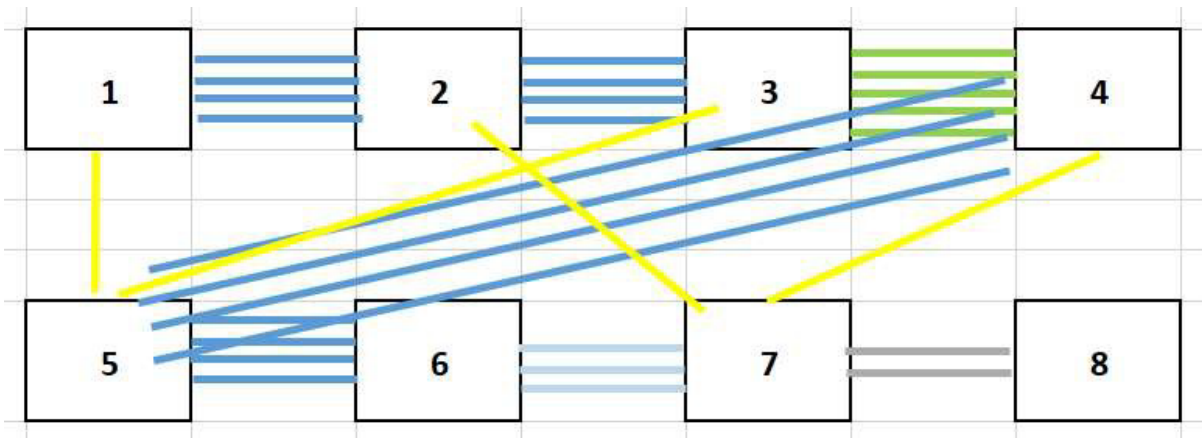


From here, I took an arbitrary placement of the departments and formed a matrix and then drew the relationships on the matrix.

DEPARTMENTS								
	1	2	3	4	5	6	7	8
1	—	20						
2	15	—	25				4	
3		5	—	40	5			
4			5	—	10			
5	1			20	—	30		
6						—	20	
7				3			—	10
8							5	—

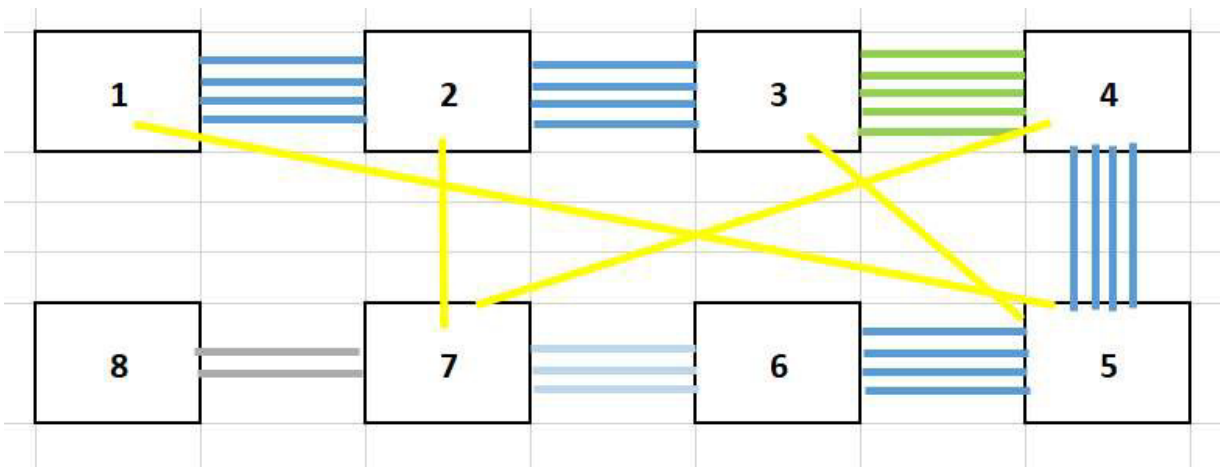


By looking at this relationship diagram, I can visually see which departments should be next to each other and which needn't be. From here, I can rearrange the blocks in the matrix to improve flow.



1	2	3	4
8	7	6	5

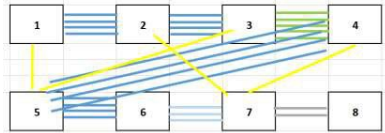
I rearranged the blocks in a new matrix that placed departments with high flow close to each other.



You can then immediately see the difference.

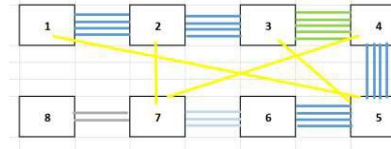
Again, this is just one solution – yours may be slightly different.

To check the improvement, we can do a rough calculation.



1	2	3	4
5	6	7	8

BEFORE	Flow	Volume	Distance	Flow x Distance
	1-2	20	10	200
	2-1	15	10	150
	2-3	25	10	250
	2-7	4	20	80
	3-2	5	10	50
	3-4	40	10	400
	3-5	5	30	150
	4-3	5	10	50
	4-5	10	40	400
	5-1	1	10	10
	5-4	20	40	800
	5-6	30	10	300
	6-7	20	10	200
	7-4	3	20	60
	7-8	10	10	100
	8-7	5	10	50
			Total:	3250



1	2	3	4
8	7	6	5

AFTER	Flow	Volume	Distance	Flow x Distance
	1-2	20	10	200
	2-1	15	10	150
	2-3	25	10	250
	2-7	4	10	40
	3-2	5	10	50
	3-4	40	10	400
	3-5	5	20	100
	4-3	5	10	50
	4-5	10	10	100
	5-1	1	40	40
	5-4	20	10	200
	5-6	30	10	300
	6-7	20	10	200
	7-4	3	30	90
	7-8	10	10	100
	8-7	5	10	50
			Total:	2320
			Improvement:	29%

Where distances are calculated as 10 from block to block and 20 moving diagonally.

TUTORIAL :

and yet another one

Activity Relationship Chart

The activity relationship chart below was drawn up to measure flow qualitatively by using the closeness relationship values among all pairs of activities and departments.

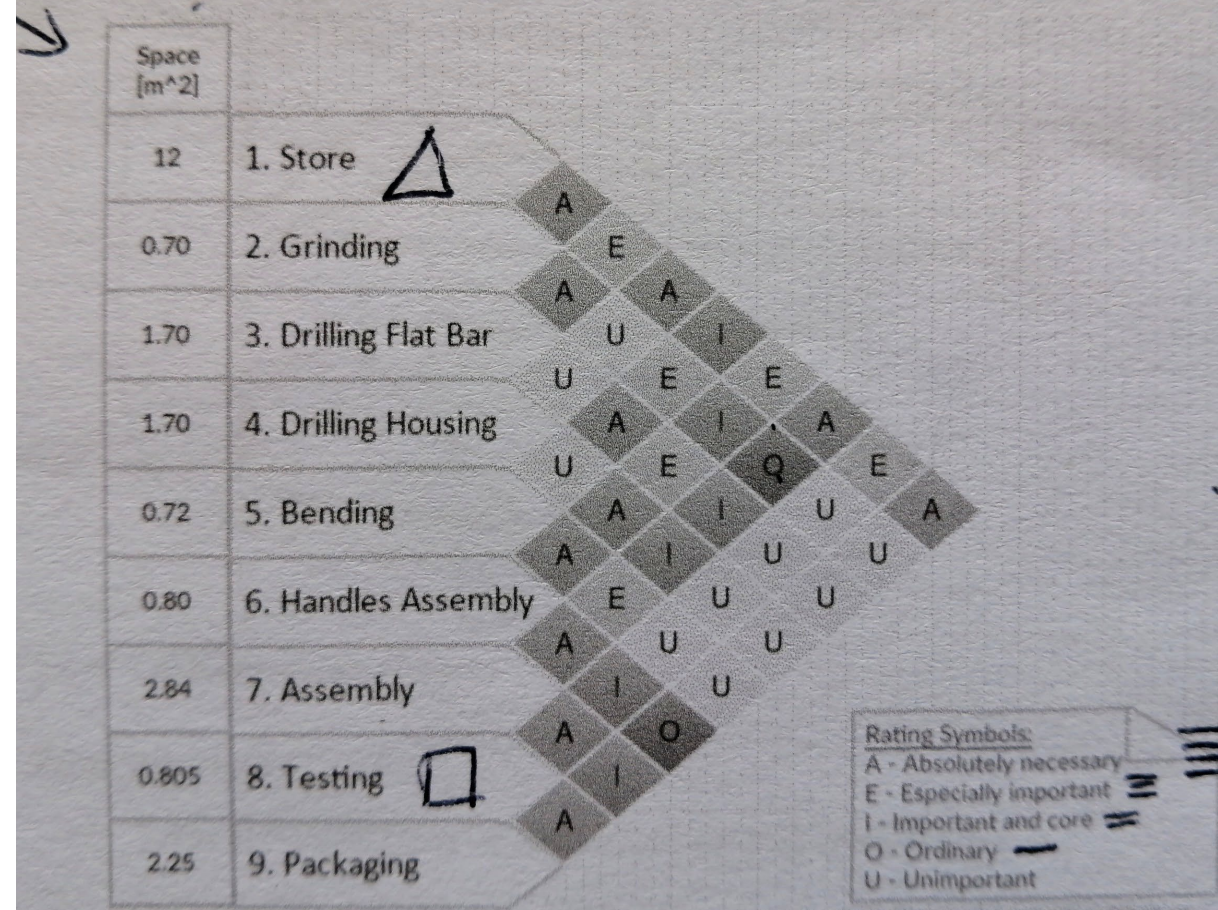
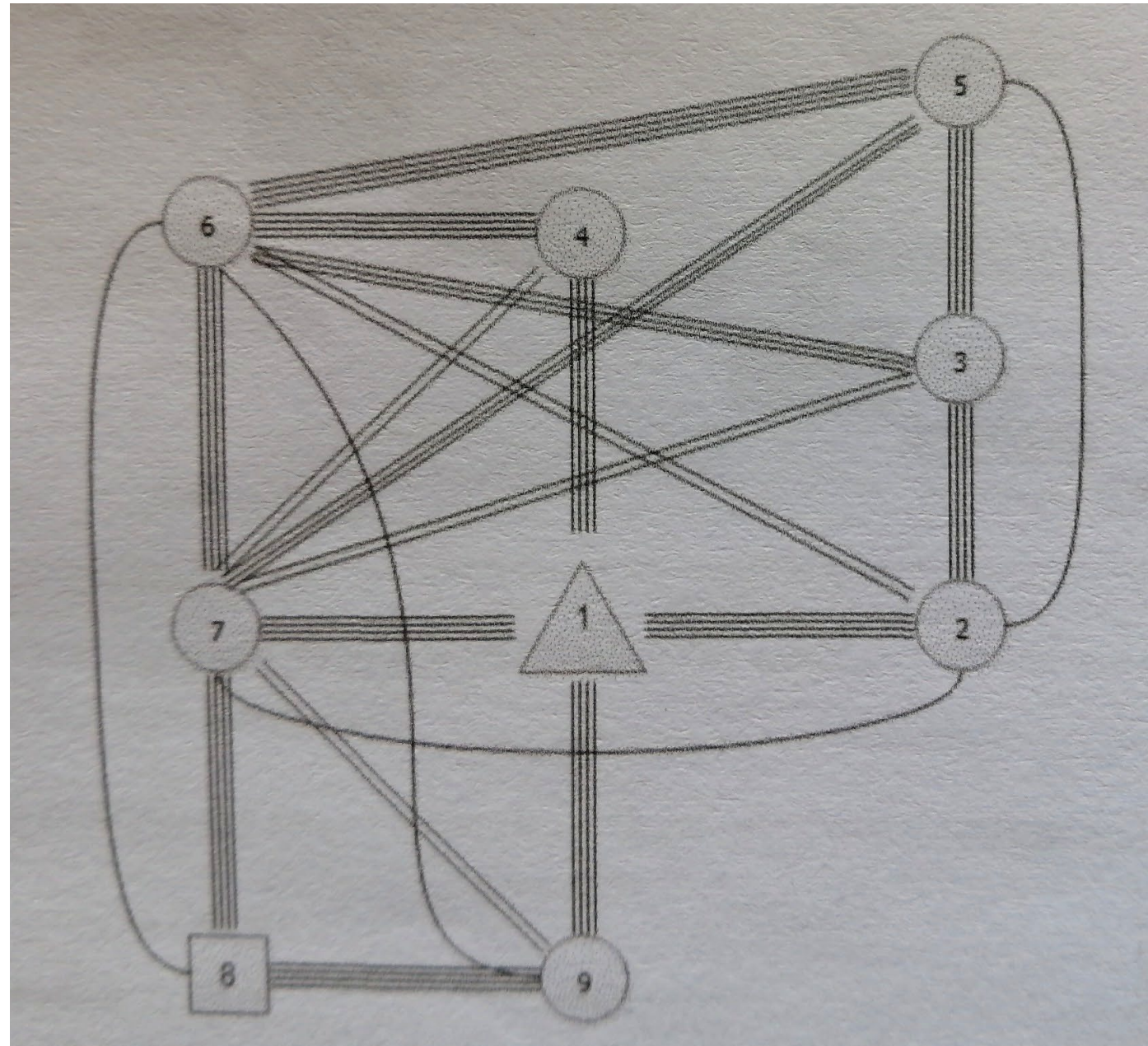


Figure 3: Activity Relationship Chart

TUTORIAL :

and yet another one



TUTORIAL :

and yet another one

Space Relationship Diagram.

